RAJYAVARDHANSINGH SHEKHAWAT
Roll Number: 23AI102
Degree Name: BTech in AI & DS
Gati Shakti Vishwavidyalaya, Vadodara

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Education

Gati Shakti Vishwavidyalaya Vadodara, Gujarat
Bachelor of Technology in AI & Data Science ;
CGPA: *8.79* August 2023 – Present

Experience

- **Summer Intern – DRM , Vadodara , Gujarat** May– June 2024
  Studied about railways engineering systems, including track circuits, block signalling, and interlocking concepts. Observed fault detection, maintenance procedures, and safety protocols used in railway operations. Gained exposure to real-time signalling coordination and operational data analysis in a safety-critical environment.

- **Summer Intern – DMRC , New Delhi** May– June 2025
  Gained hands-on exposure to metro rail control systems. Control room operations, and fail-safe design principles. Understood the role of automation and data-driven monitoring in ensuring safe and efficient metro operations.
  Contributed in real time project of optimized metro driver scheduling.

Projects

- **Crowd Analysis** August – November 2025
  Built a computer vision–based crowd analysis system to estimate crowd density and movement using Python and deep learning models. Applied image processing and data analytics techniques for safety monitoring in public spaces, with relevance to station and platform crowd management.

- **Descriptive and Diagnostic Analysis of Consecutive Train Accidents** March – April 2025
  Performed descriptive and diagnostic data analysis on historical railway accident data to identify patterns, root causes , and contributing factors. Used Python (Pandas , Matplotlib) to analyze signalling failures, human errors, and operational lapses, highlighting insights relevant to railway safety and signalling systems.

- **Maximizing Section Throughput Using AI Train Traffic Control** September 2025
  Used GenAI and Agentic AI to generate three movement plan based on live data . Station master all plans and selects the best one based on priority and feasibility. LLM-based decision engine can analyze schedules and fetches live inputs from axle counters , signal systems , and GPS to ensure safe and realistic recommendations.

- **Minor projects** August 2024 – November 2025
  Surrogate Safety Measures ,Municipal Corporation Complaint System Bank Management System , Cash Flow Mimizer, Metro Trip Charting

Key Courses Taken

Python Programming , Principles of Artificial Intelligence , Object Oriented Programming With Java , Pattern Recognition & Machine Learning , Deep Learning, Data Structure And Algorithm, Operating System , DBMS, Introduction to Rail System Engineering, Intelligent Transport System , Natural Language Processing

Skills

- **Programming Languages**: Java, C/C++, Python, SQL
- **Tools & Technologies**: MATLAB, Simulink, AutoCAD, Git, Excel
- **Operating Systems**: Windows, Linux
- **Soft Skills**: Project Coordination, Leadership, Technical Documentation & Reporting, Analytical Thinking

Achievements

- Co-Coordinator-Events – Business Club GSV, Head of Startup Cell– Hult Prize Campus Round , Head Editor of Hult 2026 magazine "The Genesis", Qualified Smart India Hackathon campus round 2025, Runner Up at Codestrom 2026